03-ACS-L-A3-S1: Calculatoare numerice II (Seria CA - 2021)

test

Due date: Tuesday, 10 May 2022, 3:00 AM
Requested files: taskX.v, test_taskX.v (Download)
Maximum number of files: 10
Type of work: Individual work

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Requested files

taskX.v

```
1  `timescale 1ns / 1ps
2  module taskX(
3      output wire o_w_out,
4      input wire i_w_a,
5      input wire i_w_b,
6      input wire i_w_sel0,
7      input wire i_w_sel1
8  );
9      wire l_w_xor;
10     wire l_w_or;
11     wire l_w_and;
12     wire l_w_nand;
13     xor(l_w_xor, i_w_a, i_w_b);
14     or(l_w_or, i_w_a, i_w_b);
15     and(l_w_and, i_w_a, i_w_b);
16     nand(l_w_nand, i_w_a, i_w_b);
17     or(o_w_out, l_w_xor, l_w_or, l_w_and, l_w_nand);
18 endmodule
```

test_taskX.v

```
`timescale 1ns / 1ps
module test_taskX;
    //Inputs
    reg l_r_a;
    reg l_r_b;
    reg l_r_sel0;
    reg l_r_sel1;

    //Outputs
    wire l_w_out;

    //local variables for loop
    integer i,j,k,r;

    //Module initialization
    taskX l_m_taskX(
        .o_w_out(l_w_out),
        .i_w_a(l_r_a),
        .i_w_b(l_r_b),
        .i_w_sel0(l_r_sel0),
        .i_w_sel1(l_r_sel1)
        );

    //Simulation tests
    initial begin
        //wave files
        $dumpfile("test.vcd");
        // dumpp all variables
        $dumpvars;
        // monitor varibles changes in values
        $monitor(
            "Time = %0t, ", $time,
            "l_w_out=%0d, ", l_w_out,
            "l_r_a=%0d, ", l_r_a,
            "l_r_b=%0d, ", l_r_b,
            "l_r_sel0=%0d, ", l_r_sel0,
            "l_r_sel1=%0d", l_r_sel1);

        #10;
        for(i=0;i<2;i=i+1)
        begin
            l_r_sel0 = i;
            for(j=0;j<2;j=j+1)
            begin
                l_r_sel1 = j;
                for(k=0;k<2;k=k+1)
                begin
                    l_r_a = k;
                    for(r=0;r<2;r=r+1)
                    begin
                        l_r_b = r;
                        #10;
                    end
                end
            end
        end
        //finish the simulation
        $finish;
    end
endmodule
```

VPL

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03-ACS-L-A3-S1-CN2-CA

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